Thứ ngày tháng năm Đỗ Quốc Tố
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Bài 1:
```
interface Mail {
    void sendMail();
    void log(); }
class MailImpl implements Mail
{ @override
  public void sendMail() {
System.out.println("Sending mail...");  }
  @override
  public void log() {
System.out.println("Log: Mail sent successfully"
}  public class Main {
  public static void main(String[] args) {
    Mail m = new MailImpl();
    m.sendMail();
    m.log();  }
}
```

Bài 2:
```
interface Shape {
    double getArea();
    double getPerimeter();
    void draw();  }
class Rectangle implements Shape {
    double width;
```

```
double height;
Rectangle (double width; double height){
  this.width = width;
  this.height = height; }
public double getArea () {
  return width * height ; }
public double getPerimeter() {
  return 2* (width +height); }
  public void draw () {
System.out.println ("Draw Rectangle"); }
} class circle implements Shape {
  double radius;
  Circle( double radius) {
  this.radius = radius; }
  public double getArea () {
  return Math.PI * radius * radius; }
  public double getPerimeter() {
  return 2 * Math.PI * radius; }
  public void draw () {
System.out.println ("Draw circle"); }
} public class Main {
  public static void main (String [] args) {
```



```
  Shape rect = new Rectangle ();
  rect.draw ();
System.out.println ("---");
Shape circle = new Circle();
  circle.draw (); }
}
```

Bài 5:

```
interface Moveable {
    void moveUp ();
    void moveDown ();
    void moveLeft ();
    void moveRight (); }
class MoveablePoint implements Moveable {
  int x, y;
  int xSpeed, ySpeed;
  MoveablePoint (int x, int y, int xSpeed, int ySpeed) {
  this.x = x;
  this.y = y;
  this.xSpeed = xSpeed;
  this.ySpeed = ySpeed; }
  public void moveUp () { y += ySpeed; }
public void moveDown () { y -= ySpeed; }
public void moveLeft () { x -= xSpeed; }
```

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```
public void moveRight() { x += xSpeed; }
public String toString() {
  return "Point (" + x + "," + y + ")"; }
} class MoveableCircle
implements Moveable {
int radius;
MoveablePoint center;
MoveableCircle(MoveablePoint center, int radius) {
this.center = center;
this.radius = radius; }
public void moveUp() { center.moveUp(); }
public void moveDown() { center.moveDown(); }
public void moveLeft() { center.moveLeft(); }
. ———— Right() { ———— Right(); }
public String toString() {
return "Circle center : " + center + " radius = " + radius; }
} public class Main {
public static void main(String[] args) {
moveablePoint p = new moveablePoint(1, 2, 1, 1);
—— Circle c = Circle(p, 5);
c.moveRight();
—— up();
```

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```
System.out.println(c); }
}
Bài 4: interface Accelerratable {
    void accelerate();
} interface Printable {
  String toString(); }
  abstract class Car {
    protected String color;
    protected double speed;
    public Car(String color, double speed) {
      this.color = color;
      this.speed = speed; }
    public String getColor() { return color; }
    — double — Speed() { — speed; }
} class BMW extends Car implements Accelerratable, Printable {
public BMW(String color, double speed) {
  super(color, speed); }
  @Override
public String toString() {
  return "BMW - color : " + color + ", Speed: " + speed; }
} class Civic extends Car implements Accelerratable, Printable {
  public Civic(String color, double speed) {
```

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```
super (color, speed); }
@ override
public void accelerate () {
  speed += 15; }
@ override
public String toString () {
  return "civic - color : " + color + ", Speed : " + speed; }
} public class Main {
public static void main (String[] args) {
Acceleratable bmw = new Bmw ("Black", 100);
_____ civic = _ civic ("white", 90);
bmw. accelerate ();
civic. ____ ();
System. out. println ((( Printable ) bmw).toString());
_____ ((( ____ ) civic). ____ ()); }
}
```

Bài 6:

```
interface I Employee {
  int calculate salary ();
  String getName ();  }
interface Reportable {
  String report ();  }
abstract class Employee implements I Employee {
```

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```
  private String name;
  ___ int payPerHour;
  public Employee (String name, int payPerHour) {
  this. name = name; this. payPerHour = payPerHour }
  public String getName () { return name; }
  ___ int getPayPerHour () { return payPerHour; }
} class PartTimeEmployee extends Employee implements
Reportable { private int hours;
  public PartTimeEmployee (String n, int p, int h) { super(n, p)
hours = h; }
  _ int calculateSalary () { return getPayPerHour () * hours }
} public class Main {
  public static void main (String [] args) {
  Reportable e1 = new PartTimeEmployee ("An", 100, 20);
  ___ e2 = _ FullTime ___ ("Binh", 200);
  System. out. println (e1. report ());
  _____ (e2. report ()); }
}
```